



SUJET STATAPP 296 - 2026-2027

Sujet proposé par : Memmi Pierre - pierre.memmi@partnersgroup.com

Organisme : Partners Group

Catégorie : Finance

Sujet :

Mandates representation and optimization via cross trades

Description du sujet :

The project focuses on analyzing real-world portfolios from multiple investors. The first objective is to develop a representation of portfolios that captures their key characteristics and relationships, enabling the discovery of non-obvious patterns, similarities, risks, and opportunities. The second objective is to optimize across portfolios by identifying mutually beneficial asset exchanges. For example, if Portfolio A and Portfolio B have different constraints, preferences, or exposures, the system could recommend swapping specific assets so that both portfolios improve their target metrics, such as expected return, risk-adjusted performance, diversification, ESG score, liquidity profile, or regulatory compliance.

Données du sujet :

Portfolio-Level Data

Portfolio identifier (anonymized)

Assets held and position weights

Portfolio market value

Historical portfolio returns

Portfolio turnover

Risk measures (volatility, VaR, Expected Shortfall)

Benchmark information

Investor type (retail, institutional, pension fund, etc., if available)

Asset-Level Data

For each asset held in the portfolios:

Asset identifier (ISIN, ticker)

Asset class (equity, bond, ETF, alternative, cash)

Sector and industry classification

Geographic exposure

Historical returns

Volatility and correlation measures

Liquidity metrics (ADV, bid-ask spread)

ESG scores

Credit ratings (for fixed income)

Factor exposures (Value, Quality, Momentum, Size, Low Volatility)

Investor Constraints and Preferences

To identify mutually beneficial exchanges, the dataset should include:

Risk tolerance limits

ESG requirements
Sector and geographic constraints
Concentration limits
Liquidity requirements
Regulatory constraints
Tax considerations (if available)
Target return objectives

Modèles envisagés du sujet :

TBC

Bibliographie du sujet :

TBC

Notation du sujet :

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Commentaires :

Il n'y a pas suffisamment d'informations sur le sujet, notamment sur les objectifs de l'étude, la disponibilité des données et, surtout, les méthodes qui seront utilisées durant le projet.